

Ques. WHAT ARE PIEZOELECTRIC SENSORS? PLEASE NAME THE DIFFERENT TYPES OF PIEZOELECTRIC SENSORS ALONG WITH THEIR SPECIFICATIONS.

Surender Kumar

Ans. A piezoelectric sensor, also known as piezoelectric transducer, is a device that uses the piezoelectric effect to measure changes in pressure, acceleration, temperature, strain or force by converting these into an electrical charge. The prefix piezo is Greek for press or squeeze. The ability of a piezoelectric material to convert a mechanical stress into electrical charge is called piezoelectric effect. Generated piezoelectricity is proportional to the pressure applied to solid piezoelectric crystal materials.

Two main sensing materials used for piezoelectric sensors are piezoelectric ceramics (such as PZT ceramic) and single-crystal materials (such as quartz). Sensitivity of ceramic materials is higher than that of natural single-crystal materials, but their high sensitivity degrades over time.

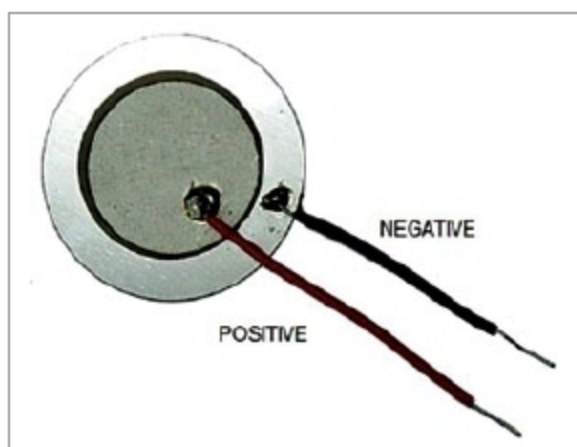


Fig. 1: Inside the quartz piezoelectric sensor



Fig. 2: Types of piezoelectric buzzers

TABLE I
SPECIFICATIONS OF PIEZOELECTRIC SENSORS

Parameters	Pin type piezoelectric	Lead type piezoelectric
Rated voltage	12V DC	12V DC
Operating voltage	3~24V DC	3~30V DC
Maximum rated current	10mA at 12V	15mA at 12V
Resonant frequency	3.5±0.5kHz	4.2±0.5kHz
Minimum sound pressure level	85dB at 12V/20cm	85dB at 12V/30cm
Pin spacing	15mm	150mm (lead length)
Outer diameter	23mm	22mm

TABLE II
MAIN DIFFERENCES BETWEEN RASPBERRY PI 3 MODEL B+ AND BEAGLEBONE BLACK (REV.C)

Parameters	Raspberry Pi 3 Model B +	BeagleBone Black (Rev.C)
CPU	Broadcom BCM2837B0, Cortex-A53 64-bit SoC @ 1.4GHz	AM3358BZCZ100, 1GHz, (2000MIPS)
GPU	Broadcom Videocore-IV	PowerVR SGX530
Memory (SDRAM)	1GB LPDDR2	512MB, DDR3L, 800MHz
Onboard network	Wi-Fi (2.4GHz and 5GHz) and gigabit ethernet	Ethernet port (10/100, RJ45)
Bluetooth support	Bluetooth 4.2, BLE	No
Video outputs	1 × full size HDMI, 3.5mm analogue audio-video jack, MIPI DSI display port, MIPI CSI camera port	Micro-HDMI (16b HDMI), 1280x1024 (maximum)
Audio outputs	3.5mm analogue audio-video jack	Via HDMI interface, stereo
Number of pins	40	92
USB ports	4x USB 2.0	1x USB 2.0

Natural single-crystal materials (quartz, gallium-phosphate, tourmaline, etc) are less sensitive but have higher stability. A commonly-available quartz piezoelectric sensor (without enclosure) is shown in Fig. 1.

There are also new single-crystal materials commercially available, such as lead-magnesium-niobate lead-titanate (PMN-PT).

There are two types of piezoelectric sensors based on the lead connection design: lead type and pin type, as shown in Fig. 2. Their specifications are listed in Table I. Both are commonly available; choice depends on size and other specifications.

Q2. WHAT ARE THE MAIN DIFFERENCES BETWEEN RASPBERRY PI 3 MODEL B+ AND BEAGLEBONE BLACK?

Preeti

A2. Both are single-board computers; however, there are many differences between the two. Main differences between the two are listed in Table II.

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender.